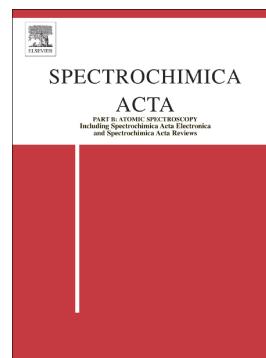


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Imaging Rare-Earth Elements in minerals by Laser-Induced Plasma Spectroscopy:
Molecular Emission and Plasma-Induced Luminescence

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Abstract

Laser-induced breakdown spectroscopy (LIBS) of atoms and ions of rare earth elements (REEs) appears to be an effective tool for REE detection and identification, specifically in imaging applications. We propose to combine this technique with the molecular emissions of LaO and YO and the plasma-induced luminescence (PIL) of REEs in luminescent matrixes. Presently, PIL is mostly sensitive to Eu, Sm, Dy, Gd, and Pr. The main advantage of the proposed technique is that both molecular emission and PIL are characterized by long plasma lifetimes, tens and hundreds of microseconds, when nearly all interfering emissions do not practically exist. Furthermore, the relatively broad emission and luminescence bands and lines enable us to use spectroscopic equipment with a relatively low spectral resolution. It is important to emphasize that the proposed experiments (atomic LIBS, molecular LIBS and PIL) are performed with the same experimental setup and from the same plasma source. Only the detection parameters (spectral range and spectrometer slit, as well as the ICCD gain, delay and width) are modified. As result, imaging of REEs becomes more sensitive and less sophisticated. As an illustration, two imaging experiments are shown to emphasize the high complementarity of these three approaches for the detection of REEs.

Keywords

**Laser-Induced Breakdown Spectroscopy (LIBS), Rare Earth Elements (REE),
Molecules, Plasma-Induced Luminescence (PIL), Imaging, Minerals**

1. Introduction

Rare earth elements (REEs) are among the main markers in geochemistry, since they are all simultaneously present in various amounts in natural minerals and thus contribute to sign genetic processes. Furthermore, they have many important applications in modern technology, and an increasing demand for these elements is straining their supply [1, 2]. To optimize their resources, there is an increasing need of analytical techniques able to rapidly detect and image their distribution in minerals [3]. Presently, photo excited luminescence, specifically laser-induced luminescence, is the preferred technique and is extremely sensitive and selective [4]. Nevertheless, several disadvantages of luminescence technique exist. First, several REEs, such as the very practically important Y and La, are not luminescent, while they both may be detected by the molecular plasma emissions of LaO and YO. Second, luminescence is only efficient with weakly doped suitable luminescent matrixes, the number of which is relatively small. For example, it is not applicable in the case of many industrial REE-bearing minerals, in which the REE levels are relatively high, because of concentration quenching.

Laser-induced breakdown spectroscopy (LIBS) is an efficient tool for the detection and identification of REEs, using the narrow emission lines of individual atoms and ions. Nevertheless, the detection of REEs by LIBS in geological samples is difficult due to the significant spectral interferences and relatively low content of REEs in the main geochemical materials (soils, sediments, minerals and rocks) [3, 4]. The first reported LIBS research on yttrium (Y) by Ishizuka [5] thoroughly studied the plasma dimensions, emission intensities and the calibration curves of several REEs in different salt matrices. As a result, the lowest limit of detection for Y was 2 ppm in a sodium chloride matrix. Harilal et al. [6] used LIBS to analyze the temporal and spatial changes of Y in a laser-induced plasma generated on the superconductor $\text{YBa}_2\text{Cu}_3\text{O}_7$, and Buckley et al. [7] detected Y in the exhaust fumes. Martin et al. [8] provided an extensive list of emission lines based on the REE samples prepared from reference materials. LIBS might be effective for the determination of light REEs, such as Y and La, under geological exploration since the achieved LODs are sufficiently lower than their Earth crustal abundances: 21 ppm for Y and 31 ppm for La [9]. Rompannen et al. [10] used not quantitative Y analysis, but the mapping of the rare-earth-bearing minerals eudialyte and catapleiite. The mapping of REEs different from La and Y, such as Eu, Nd, and Pr,

was possible in an artificially activated matrix of CaMoO_4 [11]. More recently, some of the authors reported the capability of LIBS to image light REEs in carbonate matrices and in a single-shot configuration [12]. However, this was reached at the cost of complex data processing since the spectra obtained from such samples contained multiple crowded and overlapped spectral peaks.

Apart from elemental emissions, two types of plasma-related processes might be effective for REE detection. First, the emission of radicals, formed as the atoms recombine in the cooling plasma, may have an interesting potential. This approach enables the high-sensitivity detection of halogens by their radicals with alkali-earth elements [13]. Certain REEs, such as Y and La, form, with oxygen, strongly emissive radicals, as already used in arc-induced plasma [14], and introduced for application in LIBS [3]. Second, PIL excited in the luminescent matrix by the plasma emission itself [15] can also be extremely attractive for detecting heavy REEs. In this manuscript, our aim is to demonstrate the benefit of LIP as an emission source for imaging REEs in minerals using different configurations of detection to exploit elemental, molecular and luminescence signals. These results show the high complementarity of these methods as well as the interesting versatility of laser-induced plasma, offering attractive perspectives for the detection of REEs. To the best of our knowledge, this is the first time that such experiments are conducted for elemental imaging (i.e., single-pulse configuration).

2. Experimental

2.1 Setup

The used experimental setup has been described in several papers [16-18]. Briefly, the experiment was conducted by using a Nd:YAG laser with an 8 ns pulse duration operating at 100 Hz and at the fundamental wavelength. The laser beam was focused onto the sample by a 15x magnification objective (Fig. 1a). The LIP measurements were performed at room temperature under ambient pressure in air. The laser scanning was performed line by line in raster scan mode with the use of a motorized XYZ stage. The ablation craters, observed afterward with optical microscopy, were less than 10 μm in diameter.

Two spectrometers (Shamrock 500 and Shamrock 303, Andor Technology) equipped with an intensified charge-coupled device (ICCD) cameras (iStar, Andor Technology) were used, allowing the simultaneous detection of two spectral ranges in different temporal domains. The Shamrock 500 (equipped with a 600 l/mm grating, blazed at 300 nm) was used for elemental LIBS, while the Shamrock 303 (equipped with a 1200 grating blazed at 300 nm) was used for either molecular or PIL experiments. The light emitted by the plasma was collected by two collection systems, including a quartz lens coupling the light to a round-to-linear fiber bundle connected to each spectrometer. Each fiber bundle was composed of 19 fibers, each with a 200 μm core diameter. The two ICCD cameras were synchronized with the Q-switch of the laser, and spectra were acquired in full vertical binning mode. The laser energy was stabilized throughout the experiment and was fixed to 1 mJ per pulse. Finally, homemade software developed in the LabVIEW environment was used to control the entire setup, allowing automatic sequences of any selected regions of interest with a preset lateral resolution.

As mentioned above, three different temporal windows (TW) were investigated: a delay of 1 μs and a gate width of 10 μs for elemental LIBS, a delay of 30 μs and a gate width of 100 μs for molecular LIBS and a delay of 100 μs and a gate width of 2.5 ms for plasma-induced luminescence. These parameters as well as the used configuration of associated spectrometer and detector are summarized in Table 1. Examples of spectra obtained in those three temporal windows are shown in Figure 1b. These spectra were obtained on a fluorapatite sample that was also used to determine the best detection parameters in the three cases. It is important to emphasize that atomic LIBS, molecular LIBS and PIL were accomplished with the same experimental setup and from the same plasma source. Only the detection parameters (spectral range and spectrometer slit, as well as the ICCD gain, delay and width) were modified. One may also note that because of the long integration time used for molecular LIBS and PIL experiments, ambient light may be a source of interference.

2.2 Samples

In the following, two types of results are presented. In each of these cases, the laser-induced plasma was generated continuously while scanning the sample surface. First, what we name as “reference” spectra were obtained from various materials by accumulating 1000 laser shots along several line scans. For this purpose, we used

metallic yttrium and synthetic matrices of CaMoO_4 powellite activated by different REEs, as well as rather macroscopically homogenous minerals: monazite $(\text{La,Ce,Nd})\text{PO}_4$, and xenotime YPO_4 . Second, mapping experiments were performed in the single-shot configuration and conducted on different heterogeneous minerals. Two imaging examples will be presented, the first obtained on a cerite sample, $(\text{Ce,L a,Ca})_9(\text{Mg,Fe})(\text{SiO}_4)_3(\text{HSiO}_4)_4(\text{OH})_3$, for which we detected the molecular emissions of YO and LaO, and the second from a white fluorapatite, $\text{Ca}_5(\text{PO}_4)_3\text{F}$, from Kovdor (Kola Peninsula, Russia) for which various REEs were identified by PIL.

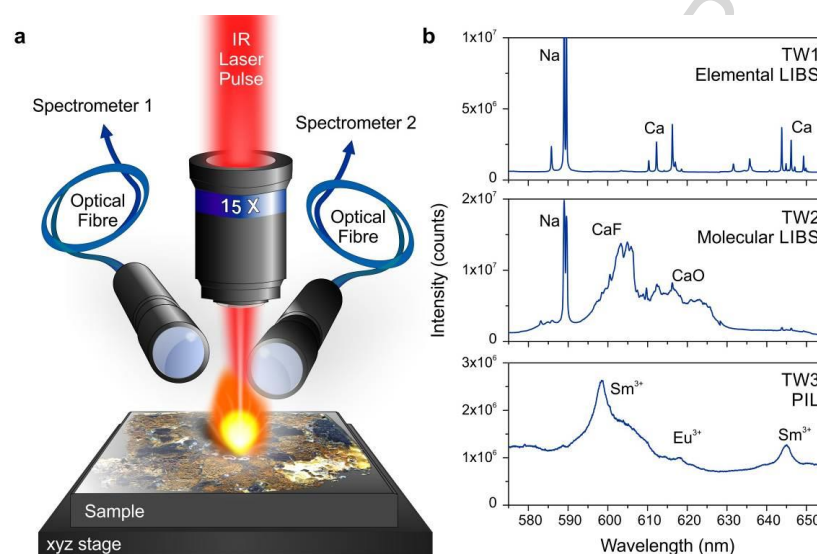


Fig. 1. a. Principle of elemental LIBS, molecular LIBS and PIL experiments. b. An example of spectra obtained from the same material but at different temporal windows (Tab. 1).

Table 1: Spectral and temporal parameters for elemental and molecular LIBS and PIL detection.

Temporal window (TW)	Delay (μs)	Gate width (μs)	ICCD gain (4096)	Spectrometer slit (μm)	Spectral resolution (nm)
Elemental LIBS (1)	1	5	0	30	0.15
Molecular LIBS (2)	30	100	2000	100	0.45
PIL (3)	100	5000	2000	100	0.45

3. Results and discussion

3.1. Molecular emission

Compared to elemental LIBS, molecular analysis by LIBS is not studied often. Molecular LIBS refers to the spectra emitted by different molecules and diatomic radicals present in the plasma, resulting from the ablation of the target or from subsequent chemical reactions with the ambient air. It seems that molecular spectroscopy with LIBS is a contradictive term since the high temperatures provided by the plasma would dissociate completely any molecular species present. Nevertheless, since the plasma evolves over time and cools down, an appropriate selection of the experimental parameters (laser energy and delay time) allows the observation of molecular emission. A notable analytical application includes the molecular emissions of halogens combined with alkali earth metals that are significantly more effective for halogen emission than are atomic and ionic lines [13]. In the case of REE detection, the use of molecular emission could be attractive for analytical applications because interfering atomic and ionic emissions are extremely weak in such a delayed plasma.

3.1.1. Spectral signature of LaO and YO molecular emissions

LaO and YO molecular emissions are very efficient in an arc-induced plasma [14] but are not used for analytical LIBS applications. Thus, at the first stage, we characterize their spectral emissions in LIP. Figure 2 presents the breakdown molecular spectra of natural LaPO_4 monazite registered with a delay a delay time of 30 μs and a gate width of 100 μs . Such a long detection delay was possible because of the long emission lifetime of these molecules. The spectra are very similar to those from an arc-induced plasma and reveal blue, yellow and red systems. Figure 3 presents the breakdown molecular spectra of metallic Y with the typical blue and orange bands of YO molecules and recorded under the same conditions previously described. It is important to note that the blue bands of YO and LaO are very efficiently produced for their detection and identification; however, unfortunately, the blue AlO emission is rather similar, and considering its major presence in several minerals, it may strongly interfere [4].

Before conducting mapping experiments, our aim was to validate whether molecular emission is sensitive enough to detect, for example, La impurities in various types of samples. These tests were first conducted on an artificial mixture of phosphate rock containing 0.5 % of La_2O_3 (Fig. 4a). Then, we analyzed a natural apatite from Kovdor (Kola peninsula) containing La at a trace level in the range of 500 ppm, a

concentration determined by inductively coupled plasma - mass spectrometry (ICP-MS) (Fig. 4b). In both cases, LaO emissions were clearly detected. Note that Figure 4 presents the red LaO system, but LaO was also detected in the yellow range. The YO molecular emission does not appear in the apatite spectrum, evidently because of the substantially lower concentration of the species (i.e., 60 ppm). In addition, no atomic or ionic emission lines of La and Y were detected in this sample.

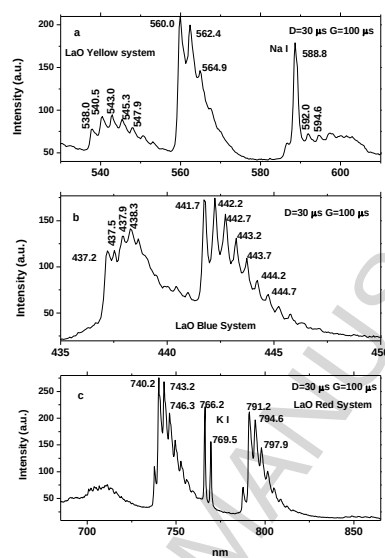


Fig. 2. LaO yellow (a), blue (b) and red (c) molecular emission systems in the laser-induced plasma of natural LaPO_4 monazite obtained under the TW2 condition (1000 shot accumulation).

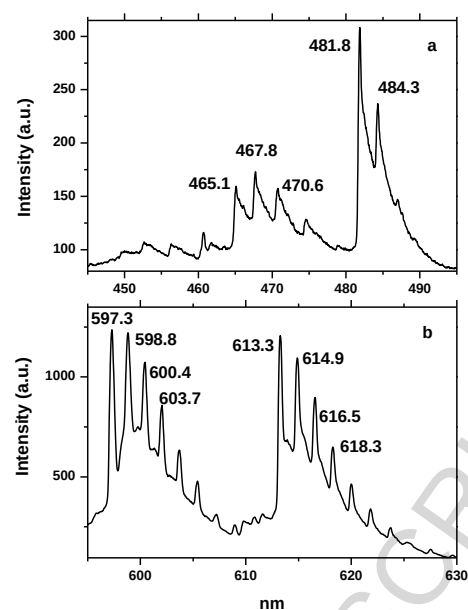


Fig. 3. YO blue (a) and orange (b) molecular emission bands by the laser-induced plasma of natural xenotime YPO_4 obtained under the TW2 condition (1000 shot accumulation).

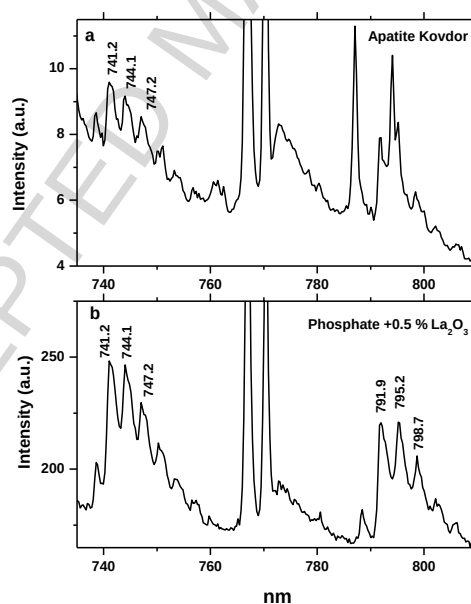


Fig. 4. Molecular LaO emissions in (a) natural fluorapatite, $\text{Ca}_5(\text{PO}_4)_3\text{F}$, (Kovdor, Kola peninsula, Russia) and (b) an artificially prepared phosphate rock (0.5 % La_2O_3) obtained under the TW2 condition (1000 shot accumulation).

3.1.2 Simultaneous molecular and elemental LIBS imaging

The imaging experiment was conducted on a cerite mineral, $(\text{Ce}, \text{La}, \text{Ca})_9(\text{Mg}, \text{Fe})(\text{SiO}_4)_3(\text{HSiO}_4)_4(\text{OH})_3$. The La concentration in this mineral is usually approximately 1.0-2.0 % [19], and both the La II ion and LaO molecule were readily detectable. The Y concentration in cerite is significantly lower, between 0.05 - 0.6 % [19], but its ionic and molecular emissions were also detectable (Fig. 5). The lateral resolution was 25 μm , and the sequence represented 302x400 pixels (720 and 400 pixels along the x- and y-axes, respectively), with one spectrum per pixel (no accumulation). The duration of this analysis was less than 1 hr. During this measurement, the two spectrometers were used simultaneously. The Shamrock 500 was configured with the TW1 detection parameters (elemental LIBS), while the Shamrock 303 was configured with the TW2 parameters (molecular LIBS) (Table 1). Figure 5a presents the experimental results. An optical image of the sample surface, as well as the main elemental images, is present at the top. These elements constitute the direct markers of the various mineral phases present. Their emission lines were selected to be unaffected by any possible interference from other lines. The heterogeneity of the sample is also highlighted by the merged image, showing a RGB (red-green-blue) representation of Fe, Ce and Si distributions. Two mineral phases related to Ce are present: one Si-bearing phase (cyan color of the merged image) and the other without Si (dark blue color). The four final images of Figure 5a represent the atomic emissions of La and Y with their corresponding molecular emissions, LaO and YO, respectively. Globally, the atomic images show a relatively good agreement with their corresponding molecular images. There are, however, some differences, which can be explained by the following. The extraction of elemental signals in such sample is very complex. The main reason is that in these types of matrices, containing metals with an extremely dense emission structure (Fe, U, Ce, Cu...), extracting the signal associated with a minor or a trace element is indeed not straightforward [12, 20] and can contribute to a severe artefact in the final results due to spectral interference. This aspect is reinforced in heterogeneous samples, such as in the present case, and the elemental images of both La and Y are probably not absolutely accurate.

In contrast, the molecular emissions have a significantly simpler emission structure that helps in the extraction of their corresponding signals. As an illustration, in Figures 5b and 5c, we show typical single-shot spectra obtained for elemental and molecular LIBS, respectively. From a spectral point of view, the elemental spectrum presents a complex structure (numerous emission lines present), in contrast with the molecular

spectrum. The long delay used in the case of molecular LIBS enables the removal of practically all atom and ion emissions (except some lines with a low excitation level). There is then no need to work with a high spectral resolution, which provides the opportunity to increase the signal without losing the spectral purity. Note that the CaF molecule is also detected in a few pixels of this sample, reflecting the F presence at the 1.5-2.0 % level [19] (Figure 5c).

The elemental and molecular LIBS images reveal heterogeneities. The cerite sample appears to be mainly composed of the Ce-rich cerite with individualized La-rich domains (dark blue phase in the merged image) revealed by La and LaO emissions. In addition to REEs, Fe, which is possibly incorporated in the cerite lattice, appears to be present in a separated Fe-Cu phase (possibly the CuFeS_2 chalcopyrite, as the red-colored phase in the merged image).

3.2. Plasma-induced luminescence (PIL)

Luminescence is a way of converting various types of energies, such as optical, nuclear, electric, mechanical, and chemical, into light emission. Many types of luminescence, named after the energy source that excites the emission of photons, are known. Plasma presents a good excitation source and PIL exists together with regular plasma emission when the excited matrix is principally capable of luminesce and contains luminescence centers. PIL is generated at early stage of plasma evolution and lasts a relatively long time, hundreds of microseconds for trivalent REE. In the case of extremely strong atomic, ionic and molecular emissions in plasma, such luminescence is difficult to detect at the beginning of plasma life and we need very long delay time, where the interfering plasma emission disappears and the luminescence remains relatively strong. It is the case for minerals studied in this paper where PIL usually dominates the emission excited by plasma after an extended delay time of more than 100 μs and with a very broad detection gate of several ms [15]. In the case of weak interference from plasma emission and very strong luminescence, it may be easily detected even at the beginning of plasma life. The examples are Eu^{2+} in fluorite and Cr^{3+} in kyanite [15].

With UV and visible excitation, lanthanide compounds have been widely reported as having the properties of exhibiting an intense narrow-band emission via the efficient intramolecular energy transfer from the ligand's excited state to the central metal ion. The characteristic emission spectra of lanthanide compounds in the visible, near

ultraviolet, and infrared attributed to transitions between $4f$ levels are due to the fact, that they present sharp lines with oscillator strengths typically on the order of 10^{-6} . It has to be noted that the REE PIL spectra are the same as in the case of traditional photoluminescence and subjected to line's broadening. The $4f-4f$ energy levels of trivalent REE are actually split by the crystal field. In fact, the splitting is relatively small due to the shielding by the $5s^2$ and $5p^6$ electrons and it is considered small for luminescence: whereas the crystal field strength in case of transition metals ions (d^n) is characteristically a few times 10000 cm^{-1} , it amounts in the trivalent REE (f^n) a few times 100 cm^{-1} . Additional lines broadening is connected to multiple structural positions of REE inside minerals structure with slightly different emission spectra. Luminescence is extremely sensitive technique for REE^{3+} because of high quantum yield of those luminescence centers. The potentially better analytical ability of PIL compared to traditional LIBS is explained by the fact that after long delay time the luminescence remains relatively strong due to long luminescence decay time of trivalent REE and possibly due to long electron recombination process during REE excitation while the interfering atomic and ionic emissions are substantially weaker [4, 15]. The long decay time of trivalent REE emission was successfully used for time-resolved luminescence mapping in order to separate it from the auto-fluorescence of the sample, which usually has short decay time of approximately 100 ns, allowing the latter decay to zero before to start the measurement of REE luminescence [21].

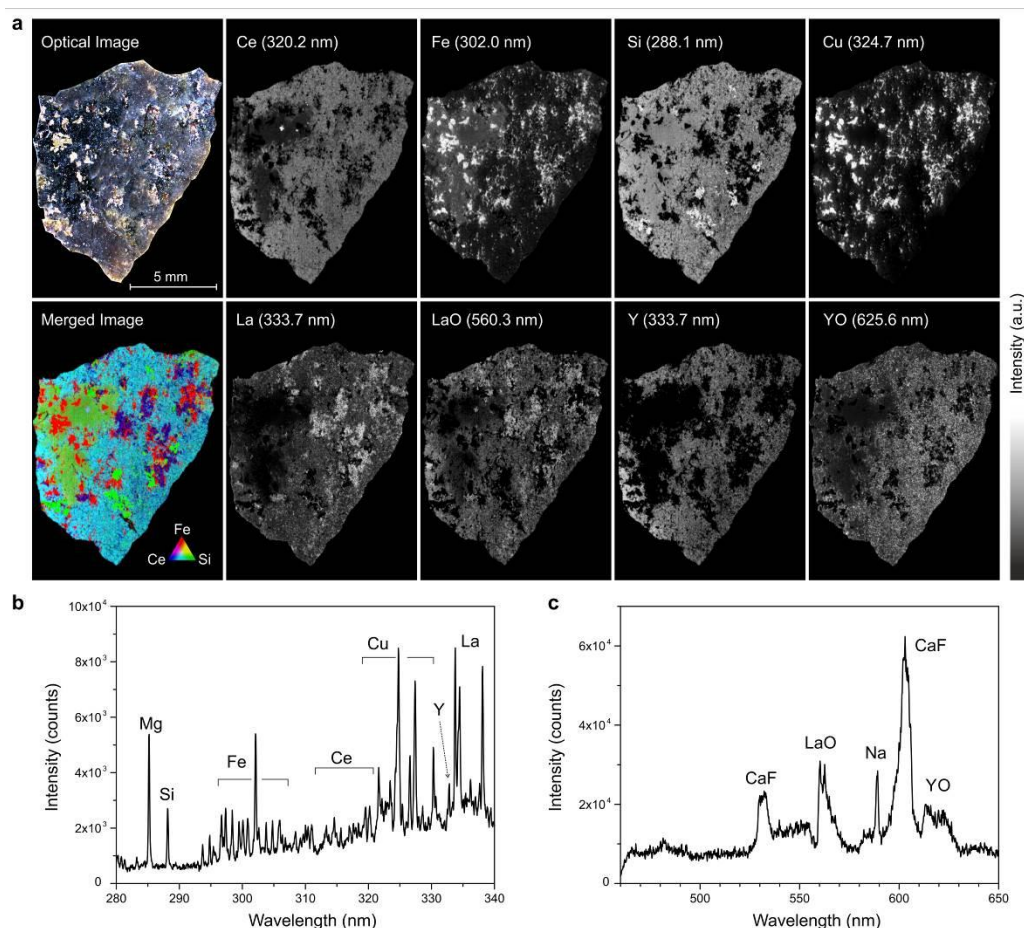


Fig. 5. Elemental and molecular LIBS imaging of a cerite sample. a) Examples of elemental and molecular images, b) A typical single-shot elemental spectrum, and c) A typical single-shot molecular emission spectrum.

3.2.1. PIL spectral signature of luminescent REE

Figure 6 presents the PIL spectra of the synthetic CaMoO_4 powellite artificially activated by 1 % Eu (Fig. 6a) and 1 % Pr (Fig. 6b). Such spectra are identical to those detected in this compound by traditional photoluminescence [22-24]. As seen, the PIL response is excellent with a signal to noise ratio (retrieved from the peak maximum) higher than 1000 in both cases. Furthermore, trace signals are observed for Pr^{3+} (Fig. 6a) and Eu^{3+} (Fig. 6b), present as impurities in such samples. Other experiments were conducted on the natural fluorapatite $\text{Ca}_5(\text{PO}_4)_3\text{F}$ from Kovdor. The results presented in Figure 7 clearly demonstrate the detection of various trivalent REEs such as Dy, Gd, Tb, Sm and Eu. In this specific apatite sample, the following range of REE concentrations is present: Tb 4 ppm, Nd 300 ppm, Dy 20 ppm, Eu 10 ppm, Gd 40 ppm and Sm 40 ppm. These concentrations are based on ICP-MS measurements. It is important to mention that no atomic or ionic lines of these elements are detected in the

LIBS configuration, which evidently shows the high sensitivity of PIL for these elements compared to that of LIBS. According to our preliminary data, the sensitivity may be similar or even better than achieved by XRF technique with synchrotron excitation where the Y, La, and Ce contents were determined with precision of about $\pm 10\%$ in the range of 100–200 ppm and more, about $\pm 15.0\%$ rel % for the content range of 50–100 ppm, and about $\pm 20.0\%$ rel% for contents less than 50 ppm [25].

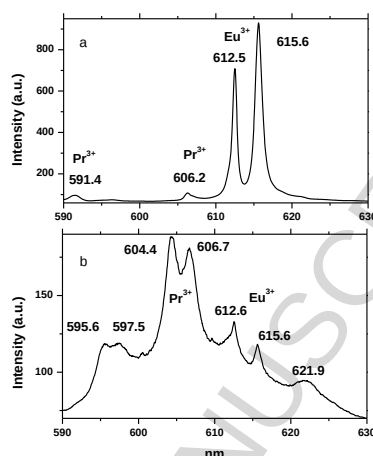


Fig. 6. Plasma-induced luminescence of (a) Eu^{3+} and (b) Pr^{3+} (each one is as 1 % content) in the synthetic CaMoO_4 powellite activated by REEs.

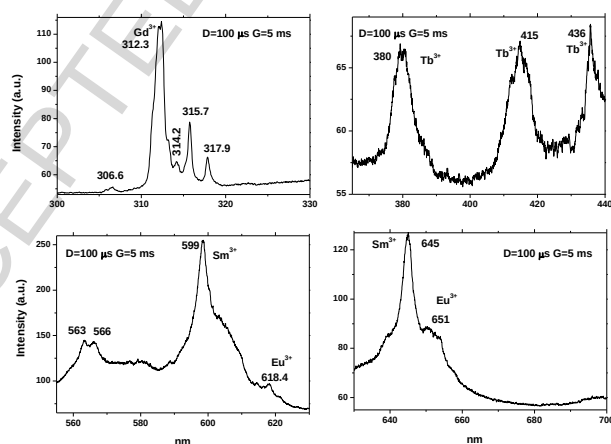


Fig. 7. Plasma-induced luminescence of Gd^{3+} , Tb^{3+} , Sm^{3+} and Eu^{3+} in the same natural magmatic fluorapatite $\text{Ca}_5(\text{PO}_4)_3\text{F}$ from Kovdor (Kola Peninsula).

3.2.2. Simultaneous PIL and LIBS imaging

The PIL imaging experiment was conducted on the Kovdor apatite with a lateral resolution of 25 μm and a mapping sequence of 280 000 pixels (560 and 500 pixels

along the x- and y-axes, respectively), again with no accumulation. As done for the molecular imaging, the two spectrometers worked simultaneously. The Shamrock 500 was configured with the TW1 detection parameters (elemental LIBS), while the shamrock 303 with the TW3 parameters (PIL) (Tab. 1). Two different scans of the same region were accomplished. The first focused on the detection of Gd^{3+} in the UV, and the other for the detection of luminescent REEs in the visible range. Figure 8a presents the corresponding results. An optical image of the sample surface, as well as the main elemental images, is present at the top of the Figure. The heterogeneity of the sample is highlighted by the merged image, showing a RGB representation of the Ca, Fe and Si distributions. The final three panels represent the luminescence images of Gd^{3+} , Sm^{3+} and Eu^{3+} . Ce^{3+} was not detected, possibly because of its extremely low decay time of 20-30 ns [3]. In addition, no elemental signal (either atomic or ionic) associated with these three elements have been detected.

It is interesting to note the degradation of the image resolution of PIL maps compared to that of elemental maps. Indeed, during PIL, the signal is not solely related to the ablated mass but to the response of the bulk material to the excitation produced by the plasma emission. The PIL resolution is then related to the properties of the collection system, and finally, to its magnification and to the dimension of the optical fiber. In our case, it is estimated to be approximately 500 μm . However, although the image definitions associated with PIL and LIBS are different, the localization of the luminescent REEs can be clearly determined to be in the white regions of the apatite sample. These regions correspond to high levels of Ca and P signals and a low level of Fe, which corresponds to the REE presence as impurities in apatite.

These results demonstrate the extreme versatility of laser-induced plasma (LIP). Such plasma is indeed a fantastic emission source, allowing the detection of the emission from ions, atoms, and molecules, as well as the luminescence response of the bulk materials. Again, we want to emphasize that all the experiments presented in this manuscript were conducted with the same instrument. Only the detection parameters of the used detectors were correspondingly adjusted for optimizing the different signals. Such versatility should open interesting perspectives for LIP in general, and in particular, for the detection of REEs. Such elements generally suffer from a poor detection limit in LIBS because of their multiple lines, weak emission and the need to use a high spectral resolution to prevent possible interference. The possibility to use molecular or luminescence detection offers new opportunities, in particular in terms of sensitivity.

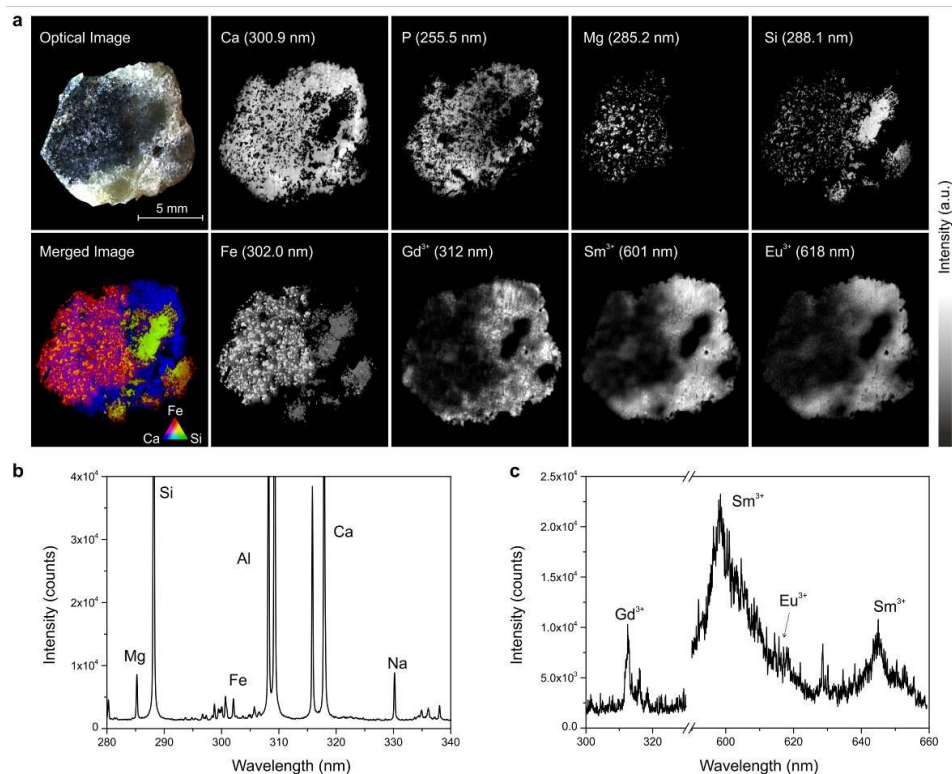


Fig. 8. Elemental and luminescence imaging of the Kovdor fluorapatite: a) elemental (Ca, P, Mg, Si, Fe) and luminescence (Gd^{3+} , Sm^{3+} , Eu^{3+}) images, b) a typical single-shot elemental LIBS spectrum, and c) a typical single-shot luminescence spectrum (PIL).

The quantitative evaluation of the detection sensitivity associated with the three approaches (i.e., elemental LIBS, molecular LIBS and PIL) is beyond the scope of this paper. We presently investigate this subject. Nevertheless, the ability to detect REE in natural objects, such as apatite, where REE present in trace amounts and not visible by traditional LIBS demonstrates the higher sensitivity of those new techniques. In addition, it is worth mentioning that the spectrometer and detector used were particularly efficient for UV detection (i.e., all the gratings used were blazed at 300 nm) and were less efficient in the visible or the near-infrared range. A more suitable spectrometer could give an additional gain of at least one order of magnitude in the visible and near-infrared ranges. This aspect may explain why Nd^{3+} , which has a strong luminescence in the near infrared, was not detected.

4. Conclusions

LIBS appears to be a very effective tool for REE detection and identification, while atomic and ionic emission enables the effective detection and identification of all REEs in their major and usually minor concentration levels. Dealing with trace amounts is more difficult because weak signals are very often not detectable because of the strong interference from major and minor elements, such as Fe, U, Ce, Ca, Cu and so on. The molecular plasma emissions of LaO and YO represent a good alternative, where long delay times of 20-100 μ s and broad emission bands enable us to conduct an experiment with very low interference and by spectroscopic devices with a low spectral resolution. In the case of trace amounts of REEs in luminescent matrixes, plasma-induced luminescence (PIL) is extremely sensitive for Eu, Sm, Gd, and Tb. REEs such as Ce, Ho, Er, Tm, and Yb are presently not identified by PIL, which will be the subject of our subsequent research.

The combination of these various laser-induced plasma spectroscopies enables the fast mapping of major, minor and trace elements in geological materials, including challenging REEs, allowing the rapid identification of chemical and mineral heterogeneities. The chemical quantification remains a challenge. Chemical quantification could be provided by microprobe or LA-ICP-MS mapping but with a weaker limit of detection for REEs and a substantially longer acquisition time (10 to 100 times longer) than that of the simultaneous elemental and molecular LIBS combined with plasma-induced luminescence.

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Highlights

- Combination of atomic, molecular emissions (LIBS) and plasma-induced luminescence (PIL) of REEs.
- Such combination was applied for mapping of REEs in heterogeneous minerals.
- All detections are performed with the same experimental setup and from the same plasma source.

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